

Q1. Define List. Write various operations that can be performed on List.

Definition:

List ek ordered and mutable (change thai sake tevo) linear data structure chhe Python ma. Ema square brackets [] no use thay chhe and alg-alg type no data (heterogeneous elements) store kari shakay chhe.

Operations on List:

- **Creation:** Navi list banavva mate:

```
Python
my_list = [10, 20, 30]
```

- **Append:** List na end ma element add karva mate:

```
Python
my_list.append(40) # Output: [10, 20, 30, 40]
```

- **Insert:** Specific index par value add karva mate:

```
Python
my_list.insert(1, 15) # Index 1 par 15 add thashe
```

- **Pop:** Last mathi athva specific index parthi item delete and return karva:

```
Python
val = my_list.pop() # Last element nikli jashe
```

- **Remove:** Value na aadhare direct item delete karva:

```
Python
my_list.remove(15) # 15 list mathi delete thai jashe
```

- **Indexing / Slicing:** Element access karva:

```
Python
print(my_list[0]) # Pehlo element malshe
```

Q2. Write an algorithm for PUSH and POP operation of stack in python.

Stack hamesha **LIFO (Last In First Out)** principle follow kare chhe.

PUSH Operation no Algorithm:

- **Step 1:** Start.
- **Step 2:** Check karo ke stack full chhe ke nahi (jo fixed size MAX hoy to).
 - Jo $TOP == MAX - 1$ hoy, to "Stack Overflow" print karo ane exit thao.
- **Step 3:** Navi value item input lo.
- **Step 4:** Python list ma `stack.append(item)` karo (athva $TOP = TOP + 1$ kari `stack[TOP] = item` set karo).
- **Step 5:** "Item pushed successfully" print karo.
- **Step 6:** Stop.

POP Operation no Algorithm:

- **Step 1:** Start.
- **Step 2:** Check karo ke stack empty chhe ke nahi.
 - Jo $len(stack) == 0$ (athva $TOP == -1$) hoy, to "Stack Underflow" print karo ane exit thao.
- **Step 3:** Python list ma `item = stack.pop()` karo (athva `item = stack[TOP]` laine $TOP = TOP - 1$ karo).
- **Step 4:** Delete kareli value (item) display karo.
- **Step 5:** Stop.

Q3. Differentiate between Stack and Queue.

Feature	Stack	Queue
Principle	LIFO (Last In First Out)	FIFO (First In First Out)
Ends	Insert ane delete banne ek j side thi thay (TOP).	Insert REAR thi thay ane delete FRONT thi thay.
Insert Operation	PUSH	ENQUEUE

Delete Operation	POP	DEQUEUE
Pointers	Ek j pointer use thay (TOP).	Be pointers use thay (FRONT ane REAR).
Example	Plates no thaglo, Browser back button.	Ticket counter ni line.

Q3 (OR). Explain concept of circular queue.

Circular Queue ek linear data structure chhe jema last position pachi direct pehli position connect thai ne ek circle banave chhe.

- **Aa concept kem aavyo?** Normal linear queue ma jyare elements delete thay, tyare agal ni jagya khali thai jay to pan REAR end par pahonchi gaya pachi navo element add nathi kari shakato (false "Queue Full" batavshe). Circular queue ma e agal ni khali jagya pachi thi use thai shake chhe.
- **Important Formulas (Array size N mate):**
 - Insert (Enqueue): $\text{REAR} = (\text{REAR} + 1) \% N$
 - Delete (Dequeue): $\text{FRONT} = (\text{FRONT} + 1) \% N$
 - Queue Full Condition: $(\text{REAR} + 1) \% N == \text{FRONT}$
 - Queue Empty Condition: $\text{FRONT} == -1 \text{ and } \text{REAR} == -1$

Q4. Define Time and space complexity.

- **Time Complexity:**

Koipan program/algorithm ne poore-pooroo execute thava mate input size (n) na hisabe ketlo computational time joiye chhe, tene **Time Complexity** kahevaay.

- **Space Complexity:**

Algorithm na execution doriyan input size (n) mujab memory (RAM) ma ketli jagya rokay chhe (fixed code space + variable data space), tene **Space Complexity** kahevaay.

Q5. Convert following equation from infix to postfix using Stack.

$$A * (B + C) - D / (E + F)$$

Conversion Table:

Read Symbol	Stack	Postfix Output
A	(empty)	A
*	*	A
(	* (	A
B	* (	A B
+	* (+	A B
C	* (+	A B C
)	* (pop +)	A B C +
-	- (pop * kem k * ni priority - thi vadhare chhe)	A B C + *
D	-	A B C + * D
/	- /	A B C + * D
(	- / (	A B C + * D
E	- / (	A B C + * D E
+	- / (+	A B C + * D E
F	- / (+	A B C + * D E F
)	- / (pop +)	A B C + * D E F +
End	(empty, pop / ane -)	A B C + * D E F + / -

Final Postfix Answer:

$\mathbf{A\ B\ C\ +\ *\ D\ E\ F\ +\ /\ -}$

Q6. Define following terms regarding Tree data structure:

1. **Forest:** Ek karta vadhare disjoint (alag-alag) trees na group ne **Forest** kahe chhe. Tree mathi root node remove karo etle niche subtrees no forest bane.
2. **Root node:** Tree hierarchy no sauthi pehlo (topmost) node jeno koi parent node nathi hoto ane in-degree 0 hoy chhe.
3. **Leaf node:** Evi node jene ek pan child node na hoy (degree = 0). Tene terminal athva external node pan kahe chhe.

Q6 (OR). Define following terms regarding Tree data structure:

1. **In-degree:** Koipan node taraf aavti incoming branches (edges) ni sankhya. Tree ma Root sivay badha node ni in-degree hamesha 1 hoy chhe.
2. **Out-degree:** Ek node mathi bahar nikalti outgoing edges ni sankhya (etle k te node na jetla direct children hoy e).
3. **Depth:** Root node thi lai ne te specific node sudhi na path ma ketli edges aave chhe te count. Root ni depth hamesha 0 hoy.

Q7. Explain class, attributes, object and class method in python with suitable example.

- **Class:** Object banava mate nu blueprint athva design template.
- **Attributes:** Class athva object ni andar data store karti variables.
- **Object:** Class no real-world instance (je memory roke chhe).
- **Class Method:** Evi method je direct class sathe jodayeli hoy (object sathe nahi). Tena mate `@classmethod` decorator use thay chhe ane pehlo parameter `cls` aave chhe.

Example Code:

Python

```
class Student:
```

```
    college = "GTU Polytechnic" # Class Attribute
```

```
    def __init__(self, name, roll_no):
```

```
        self.name = name # Instance Attribute
```

```
        self.roll_no = roll_no # Instance Attribute
```

```
    @classmethod
```

```
def change_college(cls, new_name): # Class Method
    cls.college = new_name

# Creating Object
s1 = Student("Aarav", 101)
print(s1.name, s1.roll_no, s1.college)

# Calling Class Method
Student.change_college("GTU Central")
print(s1.college)
```

Q7 (OR). Define linear and non-linear data structure and give its examples.

- **Linear Data Structure:**

Jema data elements ek pachi ek sequential order ma store ane arrange thayela hoy chhe. Aa structure ne ek j round (single pass) ma traverse kari shakay.

- **Examples:** Array/List, Stack, Queue, Linked List.

- **Non-Linear Data Structure:**

Jema data elements sequential nathi hota pan hierarchical athva interconnected (network) rite jodayela hoy chhe.

- **Examples:** Trees, Graphs.